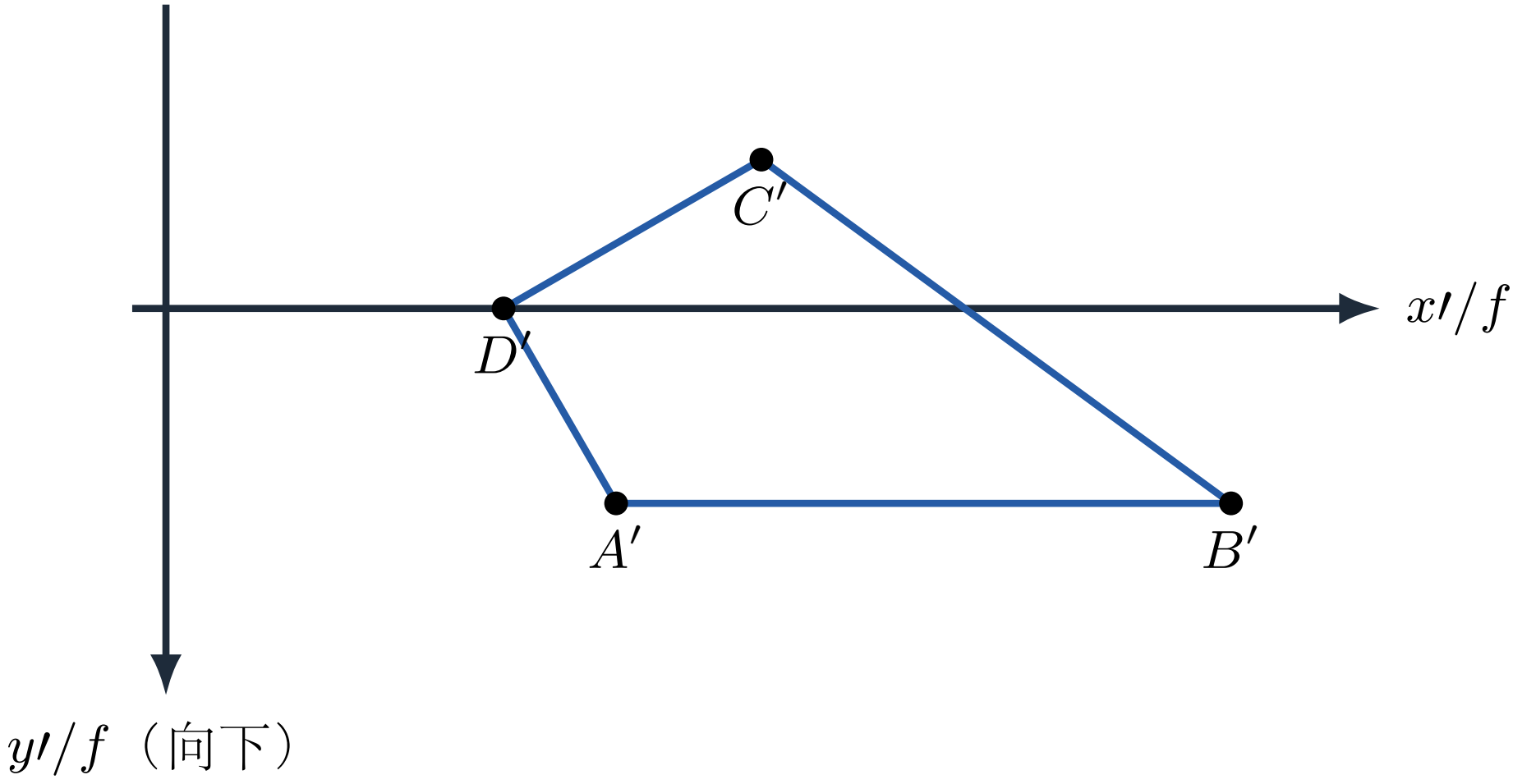
$$X\prime = 1/X,\ Y\prime = Y/X$$

$$S = (84 + 34\sqrt{3})f^2/117$$